

Answer **all** questions.

1. (a) The table shows information about some Group 1 elements. The melting point of rubidium is missing.

Element	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)
lithium	181	1 330	0.53
sodium	98	883	0.99
potassium	64	759	0.86
rubidium		688	1.53
caesium	29	671	1.93

- (i) State which of the elements **lithium**, **sodium** or **potassium**, is a liquid over the greatest temperature range. [1]

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- (ii) Estimate a value for the melting point of rubidium. [1]

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- (iii) Group 1 elements are stored in paraffin oil to prevent reactions with oxygen.

The density of paraffin oil is 0.80 g/cm³.

State which of the Group 1 elements would float on paraffin oil. [1]

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- (b) When a small piece of **sodium** is added to water, it fizzes, melts and moves around on the surface of the water.

Give **another** observation that you would expect to make when **potassium** is added to water. State the reason for the difference. [2]

Observation

Reason

- (c) Lithium reacts with oxygen to give lithium oxide.

Complete the equation for the reaction by

- giving the formula for lithium oxide
- putting numbers in the boxes to balance the equation

[2]

